Alexey Eskenazi

Software Architect • Lead Developer • Complex Systems Engineer

Saint Petersburg, Russia

Detailed career history, technical skills, patents, and selected engineering work.

My name is Alexey Eskenazi. I am a Software Architect and Lead Developer with over 25 years of experience in software engineering, project management, and complex systems architecture.

Quick Overview

Parameter	Value
Name	Alexey Eskenazi
Location	Saint Petersburg, Russia
Specialization	Software architecture and engineering for maritime navigation systems, covering both shipboard and shore-based systems

Software Patents & Intellectual Property

Note: Registered with Russian Federal Service for Intellectual Property.

Certificate No.	Project / System
2022666851	PRO-MARINER ECDIS 5000 (Electronic Chart Display and Information System)
2024681500	Pro-Mariner INS 5000 (Integrated Navigation System)
2025688320	Pro-Mariner Ice Radar 5000 (Radar Ice Imaging System)
2024681504	Pro-Mariner RADAR 5000

Featured Project

yaPDP — Interactive PDP-11/70 Environment

An open-source JavaScript PDP-11/70 emulator and browser-based environment for exploring historical operating systems and software. The focus is not only on emulating the machine, but also on making classic computing workflows approachable and reproducible.

- **Emulation:** Implements the PDP-11/70 CPU and its machine-level behavior, including traps, interrupts, and memory-mapped I/O.
- **Interactive environment:** Provides a historically oriented front-panel interface, emulated terminal and printer workflows, and browser-based access without requiring a local emulator setup.
- **Operating systems and software:** Supports prepared launch states for systems and programs such as Unix V5, 2.11BSD, RT-11, and classic games.
- **Snapshots and sharing:** Saves and restores machine states and supports sharing prepared states as reproducible entry points into an emulated system.
- **Engineering workflow:** Uses automated browser end-to-end tests and generated documentation to validate and explain complete user scenarios.
- **License:** MIT.

Launch Live Demo | **Source Code on GitHub**

Technical Skills

- **Programming Languages:** C, C++, Assembler (i80x86, MCS i8051/31), Borland C++, Watcom C++, Visual C++, MS Visual Basic, Perl, Python, Shell scripting.

- **Platforms:** IBM PC AT, Cloud VMs, Embedded Systems (Intel/ARM).
- **Operating Systems:** Ubuntu, Windows, FreeBSD, QNX, DOS.
- **Tools & Methodologies:** Microsoft Office, Visio, Project, Rational Rose, Visual SourceSafe, Subversion, Git, JIRA, Confluence, GitLab.
- **Technologies & Frameworks:** Win32 & POSIX APIs, Qt, MFC, COM, ATL, WTL, STL, Boost, TCP/IP, Docker, Kubernetes, Azure IoT Edge.

Professional Experience

2022 – Present

Evolution of Marine Digital Technologies LLC (EMDT)
Role: Principal Software Engineer / Co-owner

- Designing product architecture.
- Developing requirement specifications.
- Managing the entire software development lifecycle (SDLC).
- Conducting comprehensive source code reviews.

2018 – 2022

Wärtsilä Digital Technologies (Edge Platform Division)
Role: Principal Software Engineer / Project Manager

- Architected global enterprise products.
- Developed comprehensive technical specifications and requirements.
- Organized software release cycles and configuration management.
- Conducted code reviews for distributed teams.
- *Core Product:* Wärtsilä Edge Platform.

2005 – 2018

Transas Technologies (Navigation Products Unit)
Role: Principal Software Engineer / Project Manager

- Designed architecture for marine navigation products.
- Managed product requirements and handled stakeholder communication.
- Supervised release management and configuration control.
- *Core Products:* Hyundai Transas IBS 2.1, Navi-Conning 4000, UniNav ECDIS, etc.

2001 – 2005

Transas Software House
Role: Head of Product Management Team / Project Manager

- Managed requirements gathering and software architecture.
- Allocated resources and set up automated build environments.
- Managed version control systems and early CI pipelines.

1999 – 2001

Transas Software House
Role: Lead Software Engineer

- Analyzed customer business needs and drafted technical specifications.
- Designed software architecture and handled root-cause analysis for critical bugs.

1994 – 1999

Saint Petersburg Research Institute of Radio (NIIR)
Role: Head of Software Development Department / Software Engineer

- Developed software solutions for satellite communication systems.
- Coordinated closely with hardware and RF engineers.

1992 – 1994

MODUS LLC
Role: Software Engineer

- Developed embedded control systems for gas stations and CNC machinery.

Education

M.S. in Automated Information Processing and Control Systems (with Honors)
Saint Petersburg State Technical University, Faculty of Technical Cybernetics.

Contact

Feel free to reach out via GitHub issues/discussions or use the contact details listed in my profile.

If you need additional information or a full resume in a different format, please let me know!